Uttapreksha Patel

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AI engineer who builds agentic systems and refuses to trust a clean-looking result: built an LLM agent with tool use and memory, a production RAG pipeline, and an eval harness that passed cleanly and still missed a real hallucination I caught myself. Strong Python/PyTorch fundamentals, genuine hands-on LLM post-training depth.

Technical Skills

Agents	LangGraph, Tool Use, Multi-Step Reasoning, RAG
LLM-Post-Training	PyTorch, GRPO/PPO/DPO (from scratch), Reward Design
Evals	LLM-as-Judge, Golden-Set Design, Statistical QA, Judge Bias
Backend	FastAPI, PostgreSQL, REST APIs, Celery, Redis
Infra	Docker, Kubernetes, AWS EC2, CI/CD

Work Experience

Northeastern University | *Collaboration with Sandia National Laboratories* — **Boston, MA**
Research Assistant | Systems Engineering & Data Rigor — *February 2025 – Present*

- Scoped and shipped Python data pipelines end-to-end without a fully specified brief, processing **1M+ operation records** with automated anomaly detection; deployed to production on AWS EC2, improving pipeline reliability **30%**
- Meticulous about metrics, not just outcomes: designed controlled experiments isolating exactly which resource caused a performance regression, quantifying effects with sub-second precision across **8 identified events**
- Deployed telemetry infrastructure and diagnosed critical buffer-overflow failures through memory allocation tuning, restoring stable service under multi-node production load

TecUnique Pvt. Ltd. — **Vadodara, India**
Python Development Intern — *Dec 2023 – Apr 2024*

- Built Python REST API automation framework using Pytest achieving **93% pass rate across 30 test cases**; automated end-to-end validation with structured HTML reports and regression tracking
- Optimized SQL queries and numerical pipelines, reducing runtime **20%**; managed source control through Git in an Agile/Scrum environment

Relevant Projects

Clinical Intake Agent: Full-Stack AI Product, Catching What a Clean Pass Misses | *Python, FastAPI, Celery, Redis, React, TypeScript, Claude API, Llama, PostgreSQL, Docker, GitHub Actions | GitHub | Blog*

- Owned a 6-step production system end-to-end, solo: React/TypeScript frontend, FastAPI backend orchestrated via async Celery workers, containerized with CI on every commit
- Built a golden-set eval harness with LLM-as-judge graders; the harness reported **90.6% extraction / 0% hallucination** and passed, but manual review still caught a real failure it missed, driving a concrete fix

Full-Loop LLM Agent: Agentic Systems, Failure-Mode Evaluation | *Python, Claude API, OpenAI API, LangGraph, RAG, FAISS, FastAPI, PostgreSQL, Docker | GitHub*

- Built a LangGraph-orchestrated agent with tool use, multi-step reasoning, and adaptive memory, applying guardrails and fallback logic to recover from failures without human intervention across 75 dialogues and 15 domains
- Built an evaluation framework from scratch, LLM-as-judge graders and a 4-category failure-mode taxonomy benchmarked against 4 baselines, improving task accuracy **0.613** $\rightarrow$ **0.760** and cutting failures **64%**

RLHF Preference Data QA Pipeline: Judge Bias Measurement, Statistical Rigor | *Python, LLM-as-Judge, Statistical QA, HH-RLHF | GitHub*

- Built an automated QA pipeline for HH-RLHF preference data combining embedding-based heuristics with a local LLM judge (Qwen2.5-7B); injected 5 types of known-bad examples to measure detection, raising label-flip recall from 10% to 57% and truncation recall from 29% to 88% over heuristics alone
- Measured **32.7% position bias** in the LLM judge across 300 pairs and routed low-trust verdicts to a human review queue, raising precision from 0.44 to 0.60

Secure Code Execution Sandbox for Coding Evals: Benchmark Self-Correction | *Python, Docker, Linux Process Isolation, pass@k Evaluation | GitHub*

- Built a Docker sandbox for untrusted model-generated code with an out-of-process test oracle, closing exploits like early `exit(0)` and always-true `__eq__` that fool exit-code harnesses; evaluated Qwen2.5-Coder-1.5B on 3,280 HumanEval samples
- Found and fixed three bugs in my own harness through the study, each mislabeling failures (e.g. CPU timeouts reported as crashes); same-seed regeneration confirmed **87%** of flagged errors were genuine model mistakes, not artifacts of my own test setup

Research

Analyzing VLM-Based Approaches for Anomaly Classification and Segmentation | *Co-authored, arXiv 2026*

- Evaluated WinCLIP and AnomalyCLIP on MVTec AD and VisA benchmarks; achieved **0.916 classification AUROC** and **0.907 segmentation AUROC** through reproducible evaluation pipelines and systematic error analysis

Education

Northeastern University — **Boston, MA**
M.S. Electrical and Computer Engineering (GPA: 3.9/4.0) — *May 2026*

G H Patel College of Engineering and Technology — **Anand, India**
B.E. Electronics and Communication Engineering (GPA: 3.7/4.0) — *May 2024*